Multi-Stage Solicitation of Probability Distributions
Experiments and Theory

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Abstract

The microprediction platform built forecasting supply chains, chaining pools so that one's output became the message or the settled outcome of the next. This note is a self-critical, ex-post look at the game theory of those chains: the games were built first, and here we ask, of each, whether truthful reporting was really the best move. A chain is proper where an exogenous outcome anchors every stage; a downstream stage can then be scored on its own residual or, for the log score, at the top level. By that test the games mostly hold up, one of them only by a lucky accident.

1 The games that were built

Adam Smith's pin factory (Smith 1776) made pins cheaply by splitting the work into specialized steps. Prediction can be organized the same way. A forecast that has to be produced over and over is cheaper if it is broken into standard sub-forecasts, each contributed by whoever is best at it, the supply chain the microprediction project set out to build (Cotton 2022). Markets and contests supply one half of that, the competition that makes any single sub-forecast cheap, but not the other half: a market pays for the best forecast of its own number and stops, with no reason to pass its output to a next stage. Chaining supplies the rest. The output of one mechanism becomes the message, or the settled outcome, of the next.

We know of exactly one platform that chained probabilistic outputs this way: the retired microprediction platform, which ran several such chains at once. The only near-relative, Manifold's resolves-to-market markets, is the invalid case (§4). Chaining on point outputs is common, but that is derivative structure, not elicitation composition (§5).

The microprediction platform. A *stream* was a live quantity that someone published to the platform one value at a time — a sensor reading, a server's response time, the price of something — a real-time feed. The base game was to forecast the next value of a stream before it arrived. A forecaster submitted 225 Monte Carlo scenarios, a fixed-size sample standing for her predictive distribution, and when the value landed the pot was split in proportion to how near her scenarios fell to it. A contributor who bets to maximize her long-run wealth does best to place those scenarios according to her honest distribution: the log-optimal, all-in Kelly bet (Cotton 2022). That is a property of the log-optimal player, not of the pot split alone; a contributor optimizing something else stakes differently. This is the nearest-the-pin rule, and it only ever settled one scalar.

To make calibration and dependence into games of their own, the platform turned each of them into another scalar. Calibration first. Once the community has forecast a quantity, ask where the outcome actually landed in the forecast distribution: its percentile, or the z-score you get by pushing that percentile through a normal. Forecast well and those z-scores look standard normal; forecast badly and they drift or spread out. A z-score is just another number, so it opened its own stream, `z1~`, and predicting `z1~` meant predicting the first game's miscalibration. In the language of §2 this is the probability-integral transform run as a residual stage.

Dependence used the same idea in reverse. The joint behaviour of two streams is two-dimensional, but a pool settles one number, so the two community percentiles were folded into a single number by the Morton z-curve: interleave the binary digits of the two coordinates, the way a geohash packs latitude and longitude into one string. A pool on the folded number was a market on the copula of the pair, `z2~`; three streams folded the same way gave `z3~`. A 2020 copula contest ran exactly this on the five-minute comovements of five cryptocurrencies, contributors submitting 225 samples packed through the z-curve (Cotton 2020a). A related stacked-lottery design let competing algorithms contribute monotone calibration maps that composed into one forecast (Cotton 2020b, slides 29-31). The platform is retired.

Its successors. monteprediction is the base game with one stage repeated rather than chained. Each weekly submission is about a million joint scenarios of eleven sector-ETF returns, and the pot is split in proportion to the density a participant places on the realised vector (Cotton 2024c, 2024a); wealth threads across rounds, and the contest has run since January 2024. The MidOne contests at CrunchDAO priced residual densities directly (CrunchDAO 2024; Cotton 2024b), a residual stage without the sample smoothing.

For contrast, the point-output world. Chaining on point outputs is everywhere: derivatives, the electricity virtual bids and transmission rights on day-ahead prices (Jha and Wolak 2023), a market on another market's reported number. But it is derivative structure, not elicitation composition. And the racetrack's win versus exotic pools, index versus single-name option books, and tranche versus single-name CDS run margins and dependence in parallel books that settle independently, consistency left to arbitrage (Harville 1973; Hausch et al. 1981); parallel is not chained, and the books can disagree.

2 What can be proven

Each of these games composes stages that share one form. A *stage* is a transducer over one message type: it carries a wealth state, consumes reports that are distributional beliefs, and where it settles takes a realized outcome and returns transfers. A scoring rule, a market maker, a pool, and a parimutuel are all stages, and the operators that build and combine them, together with the convex dictionary behind them, are a companion paper (Cotton 2026a). Two facts govern whether a chain of them is valid.

A chain is proper only where an outcome anchors it. Every score is paid against a realized value or its rank; strip the outcome out and nothing pins the reports down. A chain whose final stage settles on a real outcome, with a proper score at each stage, is well founded, and truthful reporting is a stagewise equilibrium.

Proposition 1 (single-stage guarantees compose). *If each stage of a pipeline, taken with its inputs and settlement transform fixed, makes the truthful report a best response, then truthful reporting at every stage is a stagewise equilibrium: no participant gains by a deviation confined to one stage. If moreover no participant reports upstream of a stage in which they hold a position, truthfulness survives unrestricted single-stage deviations.*

Proof. With the other stages clamped, a deviation confined to stage k moves the deviator's payoff only through stage k's transfer, where the single-stage hypothesis makes truth a best response. A stage-k deviation also moves transfers strictly downstream of k; a deviator with no downstream position collects none of them. ■

This is not a Nash equilibrium of the full game: a participant who reports upstream and holds a downstream stake holds a derivative on an upstream settlement, with the usual incentive to distort the underlying (Kumar and Seppi 1992; Jarrow 1994; Hanson and Oprea 2009; Ostrovsky 2012). The platform's chains lived inside this caveat.

The residual stage, and two ways to score it. The operator that chains is the residual: a stage emits an aggregate F_1 for an outcome Y, and a second market elicits the law of the residual $U = F_1(Y)$, settling at $u = F_1(y)$. If F_1 were the truth then U is uniform (the probability integral transform), and whatever structure remains is the second stage's edge.

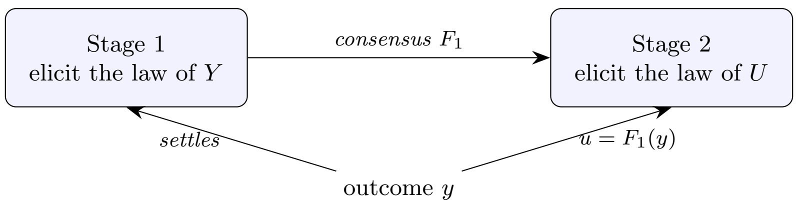

Figure 1: A residual chain. Stage 1's consensus F_1 becomes the frame for stage 2, which settles on the transformed outcome $u = F_1(y)$; the same realized y anchors both. The base pool and its z_1 calibration stream are an instance.

Proposition 2 (the residual correction is a reweighting). *For F_1 strictly increasing with density $p_1 > 0$ and a residual report with density g on $[0,1]$, the composed forecast $F = H \circ F_1$ has density $p(y) = p_1(y)\, g(F_1(y))$, so*

$$\log p(y) \;=\; \log p_1(y) + \log g(u), \qquad u = F_1(y).$$

Proof. Chain rule on $F = H \circ F_1$; take logarithms. The term $\log g(u)$ is the logarithmic score (Cotton 2026a, Thm 1) for the report g against u, strictly proper for the law of U. ■

The additive form answers a design question raised by the stacked lottery (Cotton 2020b). A contribution to the second market can be judged *locally*, by $\log g(u)$ on its own residual, or converted to a *top-level* forecast of Y and judged by $\log p(y)$. Proposition 2 makes these the same mechanism, since they differ by $\log p_1(y)$, which no downstream report can move. The equivalence is special to the log score and to scores additive under composition; for a general proper score, local and top-level scoring rank downstream contributions differently.

Dependence factors the same way. By Sklar's theorem (Sklar 1959) a joint density is $\prod_i f_i(x_i) \cdot c(F_1(x_1), \ldots, F_d(x_d))$, so

$$\log p(x) \;=\; \sum_{i=1}^{d} \log f_i(x_i) + \log c(u), \qquad u_i = F_i(x_i).$$

A pipeline of d margin stages and one rank stage settled on the rank vector pays each stage a log score of its own object, and the rank stage's report is strictly proper for the copula given correct margins — the multivariate residual, one dimension per margin.

What the pool actually elicits. The base game does not receive a density; it receives samples and smooths them. That smoothing is where propriety is won or lost.

Proposition 3 (sample elicitation; Cotton (2026c), Thms 1-2). *Score the bandwidth-h kernel density estimate of a submitted cloud by the log score. Settled at the raw outcome, the optimal cloud is drawn from a deconvolution of the belief, not the belief. Settled at the outcome jittered by the same kernel, truthful sampling is optimal, strictly so when the kernel's characteristic function is nonvanishing on a dense set.*

The correction is a jitter of the settlement, and it is the hinge on which the built games turn.

3 Were the games valid?

Take the games of §1 to the tests of §2.

The base pool: proper by lucky accident. The nearest-the-pin pool paid each contributor the density their smoothed samples placed at the realized value. By Proposition 3, scored at the raw outcome that rewards a deconvolution of the belief, not the belief: a contributor whose honest law is Gaussian would be paid most by submitting samples with variance h^2 *below* the truth. The platform escaped this, but by luck. Discrete outcomes caused computational trouble (ties when the outcome fell on a submitted sample, degenerate estimates), so the implementation jittered the settlement to smooth them over. A jitter of the outcome is the repair Proposition 3 prescribes, and it removed the deconvolution incentive — a fortunate accident, even though the amount added for numerical comfort was almost certainly not the amount matched to the smoothing scale that strict propriety asks. The hack that kept the arithmetic well behaved was the hinge that kept the game roughly honest. monteprediction settles on continuous returns, so the same accident is not available to it; whether its plug-in density split needs a fair finite-sample correction is a live question.

The `z1~` calibration stream: a residual pool that rewards sharpness. Predicting the realized z-score is a proper elicitation of its law, anchored by the exogenous outcome (with the same jitter caveat as the base pool). It is a residual stage on the base game's PIT, and it does reward sharpness. If the community's forecast was miscalibrated, its z-scores drift or spread, and whoever predicts that is paid. And if the base forecast ignored a covariate, an entrant who conditions on it forecasts the z-score's conditional law, beats the flat uniform prediction, and collects the discarded conditional information as growth, at the rate $I(R;X)$ (Cotton 2026b). What `z1~` does not deliver is a certificate from marginal statistics: a uniform PIT does not by itself witness a sharp forecast, so the premium for sharpness is claimed by an entrant who actually holds the better conditional information and stakes on it, not read off the calibration of the first game.

The `z2~`/`z3~` copula streams: proper elicitation, distorted metric. Given correct margins, folding two percentiles onto one axis and pricing the folded law elicits the copula, and for the log score the folding is a fixed bijection that changes nothing. The trouble is the *choice* of fold. The Morton z-curve is not nearness-preserving: two joint outcomes that are close on the plane can be far apart on the curve. A nearest-the-pin pool on the folded scalar therefore prices a metric that the curve, not the problem, chose, and a contributor is rewarded for placing mass near the truth *on the curve*. The elicitation is valid; the settlement metric is an artifact. The principled high-dimensional route scores random one-dimensional projections rather than one space-filling projection (Cotton 2026c), averaging a defensible metric instead of fixing an arbitrary one.

The stacked lottery: a calibration diagnostic, composed. Composing the monotone calibration maps of competing algorithms is the residual/PIT operator applied repeatedly, and it inherits the same standing as `z1~`: proper as elicitation of each stage's rank, informative as a calibration diagnostic, but not by itself a strictly proper score for the conditional forecast, since the PIT is blind to conditional sharpness.

The verdict. The chains were the right idea and mostly the right mechanism. Where they settled on an exogenous outcome and scored a rank, they were proper (the residual and copula elicitations). The base pool would have leaked an edge of order the bandwidth, but an accidental jitter introduced for discrete outcomes stood in for the settlement correction and kept it roughly honest. Where they folded a joint onto a space-filling curve, the elicitation held but the implied metric did not. A chain is as valid as its weakest anchor and its settlement transform.

4 Open problems

1. *Conservation of edge.* Wealth conservation across a chain is automatic and is not the invariant; the edge a truthful participant holds over the price, $D_G(q, \pi)$, can grow or shrink through an interface stage. For the log score the edge is $\mathrm{KL}(q\|\pi)$ and the data-processing inequality caps it (Csiszár 1967); no such inequality holds for a general Bregman edge — coarsening can *raise* the Brier edge. Open: characterize the generators and channels for which an interface cannot increase the edge.
2. *Residual markets.* A chain of residual markets is stagewise boosting with wealth as the learning rate (Proposition 2). Open: who funds each residual pot, whether a forecaster free to enter several stages withholds upstream to sell downstream, and whether the iteration converges to the conditional law. Split-conformal prediction is the one-participant degenerate case, its rent the discarded information $I(R; X)$ (Cotton 2026b).
3. *Copula settlement.* The `z2~`/`z3~` streams show that a copula elicits properly but its settlement metric is the fold's, not the problem's. Open: the right embedding for $d \geq 2$, the space-filling curves as deployed against the random projections of the point-cloud paper (Cotton 2026c), and the equilibrium when the same participants trade margin and copula stages.

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